

HEXBOARD SEQUENCER MODE QUICK MANUAL

Overview

Sequencer mode is a separate mode from Keyboard mode. It is a single-track, 32-step sequencer. Each step can hold up to 4 notes, its own note length, its own velocity, its own probability, and can play through MIDI or the onboard synth.

Enter And Exit

- From the main menu, choose Sequencer.
- In the Sequencer menu, choose Keyboard to return to normal keyboard mode.

Main Controls

- Steps: the first 8 playable keys of the first 4 rows are steps 1-32.
- Play/Stop: the top-right key starts and stops playback on a short press.
- Play/Stop also deselects the current step if one is selected.
- Performance monitor: hold the top-right Play/Stop key for 2 seconds to show audio engine load, memory, storage, and MIDI receive stats; release it to close the monitor.
- Overview: the 9th key on the second row pages through an 8-step note view.
- Overview stays lit white while idle and becomes brighter white while the overview is open.
- Blue action key: the far-right key on the second row does undo on short press and restores the selected step to the state it had when selected, including notes, length, velocity, and probability; it clears the selected step on a 1-second hold.
- Tools: the last button on the third row opens a step-tools picker for the selected step.
- Tools stays lit white while idle and becomes brighter white while the tools picker is open.
- Encoder:
 - Turn while a step is selected to change that step's length quickly.
 - Press during file or folder naming to save the name.

Step Editing

- Press a step once to select it.
- Press the same step again to deselect it.
- With a step selected, play note keys on the lower part of the board to add or remove notes from that step.
- The edit screen shows a compact info line for the selected step: L . . . % V . . . P . . . %, then the notes.
- Step notes are kept sorted from lowest to highest.
- Tap Preview can make a tapped step audition its stored notes when selected.

Step Tools Picker

- Press the last button on the third row with a step selected to open tools.
- The tools screen shows the selected step number, a compact L . . . % V . . . P . . . % line, and the current notes.
- Current tool choices are: Len, Vel, Oct+, Oct-, Prob, Tie, Copy
- The tools screen also shows: Cancel/Finished
- Len opens exact length entry for the selected step.
- Oct+ transposes the selected step up 1 octave and stays in the tools screen.
- Oct- transposes the selected step down 1 octave and stays in the tools screen.
- Vel opens exact velocity entry for the selected step.
- Prob opens exact probability entry for the selected step.

- Tie is still a placeholder entry for now.
- Copy opens target-step selection and copies the source step's notes, length, velocity, and probability into the tapped destination step.
- Cancel/Finished closes the picker when you are done.

Step Velocity

- Velocity is stored per step.
- Range is 0-127.
- Default step velocity is 96.
- Use Vel from the step tools picker to edit it.
- Velocity entry uses the encoder.
- Turn the encoder to move through: 0, 5, 10, 15, ..., 125, 127
- Press the encoder to save.
- Playback, tap preview, save, and load all use the stored step velocity.

Step Probability

- Probability is stored per step.
- Range is 0-100%.
- Default step probability is 100%.
- Use Prob from the step tools picker to edit it.
- Probability entry uses the encoder.
- Turn the encoder to move through: 0, 5, 10, 15, ..., 100
- Press the encoder to save.
- Playback, save, and load all use the stored step probability.
- Tap Preview ignores probability and still auditions the step.

Note Length

- With a step selected, turn the encoder for quick length changes.
- Use Len from the step tools picker to open exact length entry.
- Available lengths are: 0%, 25%, 50%, 75%, 100%, 150%, 200%, 250%, 300%, 350%, 400%, 500%, 600%, 700%, 800%, 900%, 1000%

Exact length screen:

- press number keys to enter the value
- the first digit typed replaces the current value
- up to 4 digits are allowed
- <- deletes the previous digit and is on the third-row first key
- CANCEL exits without saving and is on the fifth-row first key
- press the encoder to save

Playback

- Tempo is in Playback Settings and is the step clock.
- Each step is a 16th note at the current tempo.
- Hold Play/Stop for 2 seconds to show the performance monitor while held.
- Steps sets the active playback range from 1 to 32.
- Step lights beyond the active Steps count stay off, even if those steps still contain notes.
- Direction choices are: Forward, Backward, Ping-Pong, Random, Brownian, Drunk
- Play Type chooses: MIDI or OB Synth

Menu Items

- When a menu page is open, hold the bottom command button below the encoder to temporarily make the top two command buttons act like one encoder step up or down. Releasing that bottom button returns them to normal behavior.
- Save: saves to the current file.
- Save New: saves as a new file in the selected folder.
- Load: loads a saved sequence.
- Revert: reloads the last saved version of the current sequence.
- Playback Settings: Steps, Direction, Tempo, Play Type
- File Management: Rename File, Rename Folder, Create Folder, Delete File, Delete Folder, USB Backup
- Tap Preview: audition step contents when a step is tapped
- Synth Options: opens the shared synth settings page
- Update Firmware: reboots to the firmware update bootloader

Saving And Files

- Sequences are stored under /Sequences in onboard flash.
- On startup, the sequencer restores the last loaded sequence file when that remembered path is still valid.
- File Management > USB Backup starts a dedicated USB serial backup session for the / Sequences tree.
- If you back out of the USB Backup page while a session is active, the sequencer warns that leaving the page closes USB Backup and asks whether you want to continue.
- If you choose Stop Session while a session is active, the sequencer warns that stopping USB Backup ends any current transfer and asks whether you want to continue.
- While USB Backup is active, the hex buttons are temporarily disabled and their LEDs are blanked so you do not accidentally trigger other sequencer actions. The encoder still works for the menu.
- The computer-side tools are: `scripts/hexboard_backup.py` for the command line and `scripts/hexboard_backup_gui.py` for a desktop GUI.
- For skilled computer users, the simplest desktop setup is: install Python 3, install `pyserial`, then launch the GUI.
- The repo root also includes simple double-click launchers: `Launch HexBoard Backup.command` for macOS and `Launch HexBoard Backup.bat` for Windows.
- The backup tools can work on the full /Sequences tree or on a selected file or folder.
- In the desktop GUI, you can multi-select remote items with Shift-click and delete them together.
- The desktop GUI now has four action rows overall: one row beside the serial port dropdown for `Refresh Ports` and `Refresh Remote`, then three rows below for the rest of the actions.
- The rows below group: full backup actions, download and upload actions, then rename and delete actions.
- The GUI has separate `Rename File` and `Rename Folder` buttons.
- GUI renaming only allows simple sequencer-style names: letters, numbers, spaces, and `-`.
- GUI renaming rejects periods and limits names to 20 characters.
- If a rename is not allowed, the GUI shows the full rename rules, points out the broken rule(s), and then returns to the rename prompt with your previous entry still filled in.
- GUI Upload Folder always merges into the target folder on the board. It does not delete the existing remote folder first.
- During GUI Upload Folder, any existing remote file conflict is shown one file at a time so you can overwrite, skip that file, or cancel the folder upload.
- GUI Upload File warns before replacing an existing remote file, whether that file was selected directly or is already inside the selected target folder.

- The warning also tells you to select a folder first if you want to copy the file into that folder instead of replacing a selected file.
- When multiple remote items are selected, single-target actions like Download, Upload, and Rename are disabled until only one item is selected again.
- The GUI remembers the last local folder used for Upload File and Upload Folder, and it also remembers the last backup folder used for Restore Backup. Those locations return after reopening the desktop tool.
- The top GUI status line shows active operations, including Refresh Remote.
- The `Size Bytes` column shows raw file sizes in bytes.
- While that session is active, Save, Revert, and file-browser actions are blocked until the session is stopped.
- File browsers show up to 8 file/folder rows at a time.
- `^ more ^` and `v more v` indicate earlier or later pages.
- Scrolling past the first or last real file row pages automatically.
- The menu title shows the current file name as: Seq-FILENAME
- If there are unsaved changes, it becomes: Seq-*FILENAME
- If no file is loaded, the title just says: Sequencer

Naming Mode

When naming a file or folder, the board shows this entry layout:

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z SPC - 1 2 3 4 5 <- 6 7 8 9 0 CANCEL

- Press keys to enter characters.
- SPC adds a space.
- <- deletes the previous character.
- CANCEL exits naming without saving.
- digits 0-9 are available directly on the bottom two rows.
- Press the encoder to save the name.
- the physical CANCEL key is placed lower on the board to reduce accidental exits

Status And Display

- The selected step blinks.
- The overview screen shows 8 steps at a time.
- Empty steps are dark when idle.
- The performance monitor shows: AudioEng, Mem, FS, MIDI Q, Drop, and Late while the Play/Stop key is held.
- Active sequencer overlays stay awake while you are using them.
- The top title shows the loaded file name and unsaved state.